IPL Performance & Match Analytics

End-to-end data analytics project | Python • PostgreSQL • Power BI

1. Executive summary

This project analyzes 1,169 IPL matches and 278,202 cleaned ball-by-ball records across 18 seasons, from 2008-04-18 to 2025-06-03. The workflow covers data validation, feature engineering, team and player performance, toss analysis, innings scoring phases, SQL analytics, and Power BI reporting.

2. Data sources and scope

The supplied matches file contains one record per match, while the deliveries file contains ball-by-ball records. The two tables are joined by match_id. The analysis preserves team names as recorded in each season and does not merge franchise rebrands without a documented mapping.

Dataset check	Result
Match records	1,169
Delivery records (original)	278,205
Delivery records (cleaned)	278,202
Exact duplicate delivery rows removed	3
Unique seasons	18
Recorded team/franchise labels	19
Matches with winner recorded	1,146
Tied matches	15
No-result matches	8
Unmatched delivery match IDs	0
Matches without delivery records	0

3. Data preparation

Three exact duplicate delivery rows were removed. Date fields were parsed as dates, columns standardized to snake_case, and numeric fields converted to numeric types. Derived delivery fields include total_runs, legal-ball indicator, boundary flags, wicket event, bowler-credited wicket, bowler_runs_conceded, and innings phase. Wides and no-balls are excluded from legal balls; wides are excluded from batter balls faced.

4. Key findings from the supplied dataset

• Most wins by recorded team label: Mumbai Indians (151 wins).

• Highest aggregate batter runs: V Kohli (8,671 runs).

• Most bowler-credited wickets: YS Chahal (221 wickets).

• Toss winner also won 51.57% of matches where a winner was recorded.

• Phase run rates: Powerplay 7.97, Middle 7.87, Death 10.02 runs per over.

5. Analytical definitions

Batting runs: Sum of batsman_runs. **Balls faced:** deliveries excluding wides; no-balls count as faced balls. **Strike rate:** runs / balls faced × 100. **Legal balls:** deliveries excluding wides and no-balls. **Bowler-credited wickets:** dismissals excluding run out, retired hurt, obstructing the field and retired out. **Economy:** bowler runs conceded / legal overs. **Innings phase:** Powerplay overs 0–5, Middle overs 6–14, Death overs 15 onward.

6. Limitations and responsible interpretation

The 23 matches without a recorded winner comprise 15 ties and 8 no-results. Team naming changed over IPL history; results are shown under recorded names rather than merged aliases. Player leaderboards should be filtered by minimum opportunities when comparing strike rates or economy. Bowling wicket attribution follows the stated exclusions and may differ from a provider's official scorecard conventions. Toss analysis is observational and does not establish causality. The project is descriptive analytics, not a predictive model.

7. Business questions answered

Which recorded team labels have the most wins? Who leads in batting runs and bowler-credited wickets? How do run rates differ across innings phases? Does the toss-winning team also win the match? Which seasons and venues account for the highest match volumes? These questions are implemented in the Python notebook and PostgreSQL query file.

8. Reproducibility

Run Python/ipl_analysis.ipynb from the Python folder. Import Dataset/matches_cleaned.csv and Dataset/deliveries_cleaned.csv into PostgreSQL tables matches_cleaned and deliveries_cleaned. Execute SQL/ipl_analysis.sql. Load the two cleaned CSVs into Power BI, create the one-to-many match_id relationship, and apply PowerBI/measures.dax and the dashboard build guide.

Dataset attribution: cite the original Kaggle dataset author, URL and license in the public repository README. Do not claim the source data as original work.